

6(a)	any three from: 1 need to open to obtain carbon dioxide for, photosynthesis / Calvin cycle ; 2 need to close to prevent water loss by, transpiration / described ; 3 can close, at night / when dark, (to save water) as no photosynthesis ; AW 4 must close to prevent water loss when, high light intensity / high temperatures / high wind speed / water stress / drought conditions ;	3
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Question	Answer	Marks
6(b)	any six from: 1 H^+, pumped / moved by active transport, out (of guard cells) ; 2 decrease in H^+ inside (of guard cells) / proton gradient set up / AW ; 3 K^+ channels open / K^+ enters (guard cells) ; 4 water potential (in guard cells), decreases / lowers / becomes more negative ; 5 water enters (guard cells) by, osmosis / down the water potential gradient ; 6 cells, expand / become (more) turgid ; 7 outer wall thinner to allow (more), stretching / bending or inner wall thicker to allow less, stretching / bending ; 8 AVP ; e.g. light activates, ATP production / proton pump sentry of, chloride ions / Cl^- (in addition to K^+) / entry of water via aquaporins <i>max 3 if stomatal closure described</i>	6